

TRIAD: T-Cell Immunogenicity Score Reference

Comprehensive documentation of the **T-Cell Immunogenicity Scoring Model** implemented in the TRIAD platform, covering theoretical foundations, mathematical formulations, parameter tables, length adaptation, and clinical candidate ranking.

1. Scientific Origin & Core Concept

- **Reference:**

Calis JJ, Maybeno M, Greenbaum JA, Weiskopf D, De Silva AD, Sette A, Keşmir C, Peters B.
Properties of MHC Class I Presented Peptides That Inspire Immunogenicity.
PLoS Computational Biology (2013), 9(10): e1003266. [doi:10.1371/journal.pcbi.1003266](https://doi.org/10.1371/journal.pcbi.1003266)

- **Problem Addressed:**

Conventional epitope prediction pipelines (such as NetMHCpan) predict **MHC-I binding affinity and presentation** (IC₅₀ or %Rank_EL). However, **over 50% of peptides that bind strongly to MHC fail to trigger a CD8+ T-cell response in vivo.**

- **Underlying Mechanism:**

MHC-presented peptides must interact productively with the **T-cell receptor (TCR)**. TCR activation is primarily dictated by the **physicochemical properties of amino acids exposed at TCR-facing contact positions** (the solvent-exposed central bulge of the peptide), rather than the anchor residues buried inside the MHC cleft.

2. Mathematical Formulation

The raw immunogenicity score is computed as a position-weighted sum of amino acid log-odds enrichment values, excluding anchor positions buried within MHC pockets:

$$\text{Immunogenicity_Score} = \sum_{i \notin \text{Mask}} w_i \times s(aa_i)$$

Where: * i : 0-indexed position along the peptide sequence ($0 \dots L - 1$). * Mask: Set of anchor positions buried inside the MHC binding cleft (interacting with MHC, not TCR). * w_i : Positional contribution weight reflecting how prominently position i protrudes toward the TCR CDR3 loops. * $s(aa_i)$: Physicochemical log-odds enrichment score for amino acid aa_i .

3. Amino Acid Propensity Scale ($s(aa)$)

Calis et al. derived these values by comparing thousands of experimentally verified immunogenic vs. non-immunogenic pMHC complexes from the Immune Epitope Database (IEDB):

Category	Amino Acid	1-Letter Code	Log-Odds Weight (s)	Structural & Biophysical Rationale
Highly Immunogenic	Tryptophan	W	+0.719	Bulky, aromatic, and rigid aliphatic side-chains protrude deeply into the TCR CDR3 binding pocket, establishing strong van der Waals contacts and hydrophobic interactions. Negatively charged glutamate (E) readily forms salt bridges.
	Isoleucine	I	+0.432	
	Phenylalanine	F	+0.380	
	Glutamate	E	+0.325	
Moderately Positive	Arginine	R	+0.168	Favorable to neutral contributions without significant steric repulsion.
	Valine	V	+0.134	
	Alanine	A	+0.127	
	Threonine	T	+0.126	
	Glycine	G	+0.110	
	Histidine	H	+0.105	
	Aspartate	D	+0.072	
Weak / Neutral	Tyrosine	Y	-0.012	Minimal net contribution to T-cell receptor engagement.
	Asparagine	N	-0.021	
	Leucine	L	-0.036	
	Proline	P	-0.036	
Unfavorable / Disfavored	Cysteine	C	-0.175	Small, overly flexible, or highly hydrated hydrophilic residues. They fail to provide sufficient desolvation free energy or steric engagement needed to trigger TCR conformational activation.
	Glutamine	Q	-0.376	
	Serine	S	-0.537	
	Methionine	M	-0.570	
	Lysine	K	-0.700	

4. Positional Weighting (w_i) & Length Adaptation

Standard 9-mers

For standard 9-mer peptides, positional weights follow the canonical structural protrusion profile:

```
# Defined in netmhcpn_platform/immunogenicity_scorer.py
POSITION_WEIGHTS = [0.00, 0.00, 0.10, 0.31, 0.30, 0.29, 0.26, 0.18, 0.00]
#           P1      P2      P3      P4      P5      P6      P7      P8      P9
```

- **P1 & P2** ($w = 0.00$): Buried in the N-terminal A/B pockets of the MHC-I groove.
- **P4, P5, P6, P7** ($w = 0.26 \sim 0.31$): The apex of the peptide bulge facing directly into the CDR3 α and CDR3 β loops.
- **P9** ($w = 0.00$): C-terminal anchor buried in the F pocket.

Longer Peptides (> 9-mers: 10-mer, 11-mer, etc.)

Because the MHC Class I binding groove is closed at both ends, extra residues are forced into a **central zigzag arch**. TRIAD dynamically adapts by inserting a weight of `0.30` for each additional residue into the central bulge:

```
if peplen > 9:
    weights = POSITION_WEIGHTS[:5] + [0.30] * (peplen - 9) + POSITION_WEIGHTS[5:]
```

5. Allele-Specific Anchor Masking (`ALLELE_POSITION_MASK`)

Different HLA alleles utilize distinct internal binding pockets (Pockets A through F). TRIAD dynamically masks non-accessible anchor positions depending on the presenting allele:

Presenting Allele	Masked Positions (1-indexed)	Anchor Pocket Mechanism
Default / Canonical (e.g., HLA-A*02:01, HLA-A*03:01)	1, 2, C-term	Standard N-terminal P2 and C-terminal P9 anchors.
HLA-B*08:01	2, 5, 9	Pocket D anchors position P5 internally; masked from TCR contact.
HLA-A*24:02 / HLA-A*23:01	2, 7, 9	Auxiliary anchor at P7.
HLA-B*27:05 / HLA-B*44:02 / HLA-B*44:03	2, 3, 9	Pocket B/D anchors P3 as a major secondary anchor.
HLA-A*01:01	2, 3, 9	P3 aspartate/glutamate anchor preference.

6. Score Interpretation & Platform Integration

1. Raw `Immunogenicity_Score`

- Typical range: -0.4000 to $+0.4000$.
- > 0.00 : **Positive immunogenic propensity** (enriched in bulky/aromatic residues at contact sites; CD8+ T-cell activation potential).
- < 0.00 : **Low immunogenicity likelihood** (high risk of being a non-activating MHC binder).

2. Normalized S_{imm} ([0.0, 1.0])

For multi-criteria decision making, TRIAD rescales the raw score into a normalized interval [0, 1]:

$$S_{\text{imm}} = \text{clip}\left(\frac{\text{Immunogenicity_Score} + 0.4}{0.8}, 0.0, 1.0\right)$$

3. Integration into the TRIAD Composite_Score

TRIAD synthesizes four complementary biological dimensions into a single golden index to rank epitope candidates:

$$\text{Composite_Score} = \underbrace{0.40 \times S_{\text{pres}}}_{\text{MHC-I Presentation (\%Rank)}} + \underbrace{0.25 \times S_{\text{aff}}}_{\text{Binding Affinity (IC}_{50})} + \underbrace{0.25 \times S_{\text{imm}}}_{\text{TCR Immunogenicity}} + \underbrace{0.10 \times S_{\text{breadth}}}_{\text{HLA Promiscuity}}$$

7. Implementation Files in TRIAD

- Core Scorer: netmhcpa_platform/immunogenicity_scorer.py
- Batch Execution & Composite Ranking: netmhcpa_platform/batch_runner.py
- Interactive Web UI Dashboard: netmhcpa_platform/templates/index.html